

Geometria Analítica Lista 7 Jeann Victor Batista

① Equações de reta

a) $A = (3, 5, 1)$ e $B = (-2, 3, 2)$

$$\vec{AB} = B - A = (-2, 3, 2) - (3, 5, 1) = (-5, -2, 1)$$

$$\vec{AX} = \lambda \vec{AB} \rightarrow X = (3, 5, 1) + (\lambda(-5, -2, 1))$$

$$x = 3 - 5\lambda$$

$$y = 5 - 2\lambda$$

$$z = 1 + \lambda$$

Paramétrica

$$\pi: \frac{x-3}{-5} = \frac{y-5}{-2} = \frac{z-1}{1}$$

Simétrica

b) $A = (0, 1, 0)$ e $B = (1, 0, 0)$

$$\vec{AB} = B - A = (1, 0, 0) - (0, 1, 0) = (1, -1, 0)$$

$$\vec{AX} = \lambda \vec{AB} \rightarrow X = (0, 1, 0) + (\lambda(1, -1, 0))$$

$$x = \lambda$$

$$y = 1 - \lambda$$

$$z = 0$$

Paramétrica

$$\pi: \frac{x-0}{1} = \frac{y-1}{-1} = \frac{z-0}{0}$$

Simétrica, Não é possível

c) $A = (0, 1, 1)$ e $B = (0, 0, 0)$

$$\vec{AB} = B - A = (0, 0, 0) - (0, 1, 1) = (0, -1, -1)$$

$$\vec{AX} = \lambda \vec{AB} \rightarrow X = (0, 1, 1) + (\lambda(0, -1, -1))$$

$$x = 0$$

$$y = 1 - \lambda$$

$$z = 1 - \lambda$$

Paramétrica

$$\pi: \frac{x-0}{0} = \frac{y-1}{-1} = \frac{z-1}{-1}$$

d)

$A = (3, 2, 1)$ e $B = (6, 1, 3)$

$$\vec{AB} = B - A = (6, 1, 3) - (3, 2, 1) = (3, -1, 2)$$

$$\vec{AX} = \lambda \vec{AB} \rightarrow X = (3, 2, 1) + (\lambda(3, -1, 2)) \rightarrow X = (3, 2, 1) + (3\lambda, -\lambda, 2\lambda)$$

$$x = 3 + 3\lambda$$

$$y = 2 - \lambda$$

$$z = 1 + 2\lambda$$

Paramétrica

$$\pi: \frac{x-3}{3} = \frac{y-2}{-1} = \frac{z-1}{2}$$

tilibra

$$(2) a) AX = \lambda AB \rightarrow X = A + \lambda(AB)$$

$$X = 1 - \lambda \quad X = (1, 0, 2) + \lambda(-1, 1, 2)$$

$$Y = \lambda \quad \text{Logo } A = (1, 0, 2)$$

$$Z = 2 + 2\lambda \quad AB = B - A = (-1, 1, 2) = (X, Y, Z) - (1, 0, 2)$$

$$(-1, 1, 2) = (1, 0, 2) - (X, Y, Z)$$

$$A = (1, 0, 2) / B = (0, 1, 6) / AB = (-1, 1, 2) \quad (0, 1, 6) = (X, Y, Z) \rightarrow B = (0, 1, 6)$$

$$BA = A - B = (1, 0, 2) - (0, 1, 6) = (1, -1, -2)$$

$$Y = B + \lambda(BA)$$

$$Y = (0, 1, 6) + \lambda(1, -1, -2)$$

$$b) X = 1 - \lambda \quad 1 = 1 - \lambda \quad 1 = 1 - 3 \quad X$$

$$Y = \lambda \rightarrow 3 = \lambda \rightarrow 3 = \lambda$$

$$Z = 2 + 2\lambda \quad -3 = 2 + 2\lambda \quad -3 = 2 + 6 \quad X$$

$P = (-1, 3, -3)$ não pertence a reta r

$$X = 1 - \lambda \quad -3 = 1 - \lambda \quad -3 = 1 - 2 \quad \checkmark$$

$$Y = \lambda \rightarrow 2 = \lambda \quad 2 = \lambda \quad \checkmark$$

$$Z = 2 + 2\lambda \quad 12 = 2 + 2\lambda \quad 12 = 2 + 2 \times 2 \quad \checkmark$$

Q pertence a reta r .

$$(c) Z = C + \lambda AB \quad Z = C + \lambda BA$$

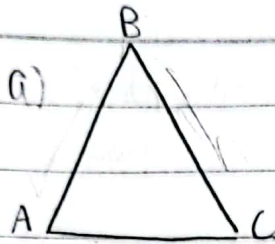
$$Z = (1, 2, -7) + \lambda(-1, 1, 2) \quad Z = (1, 2, -7) + \lambda(1, -1, -2)$$

$$X = 1 - \lambda \quad X = 1 + \lambda$$

$$Y = 2 + \lambda \quad Y = 2 - \lambda$$

$$Z = -7 + 2\lambda \quad Z = -7 - 2\lambda$$

3



a)

$$AB = B - A = (-5, 2, 3) - (3, 6, -7) = (-8, -4, 10)$$

$$BC = C - B = (2, -7, -6) - (-5, 2, 3) = (7, -9, -9)$$

$$AC = C - A = (2, -7, -6) - (3, 6, -7) = (-1, -13, 1)$$

$$r: AX = \lambda AB \rightarrow X = (3, 6, -7) + \lambda(-8, -4, 10)$$

$$X = (3, 6, -7) + (-8\lambda, -4\lambda, 10\lambda)$$

$$x = 3 - 8\lambda$$

$$y = 6 - 4\lambda$$

$$z = -7 + 10\lambda$$

$$y = 6 - 4\lambda$$

$$-7 = 6 - 4\lambda$$

$$13 = -4\lambda \rightarrow \lambda = -13/4$$

$$z = -7 + 10\lambda$$

$$-6 = -7 + 10\lambda$$

$$1 = 10\lambda \rightarrow \lambda = 1/10$$

C não está em $\overrightarrow{AB}$

$$r: AX = \mu AC \rightarrow X = (3, 6, -7) + \mu(-1, -13, 1)$$

$$d) x = 3 + \mu$$

$$-5 = 3 + \mu$$

$$-8 = \mu$$

$$y = 6 - 13\mu$$

$$\rightarrow$$

$$2 = 6 - 13\mu$$

$$-4 = -13\mu$$

$$z = -7 + \mu$$

$$3 = -7 + \mu$$

$$10 = \mu$$

B não está em $\overrightarrow{AC}$

$$r: BX = \lambda BC \rightarrow X = (-5, 2, 3) + \lambda(7, -9, -9)$$

$$x = -5 + 7\lambda$$

$$1 = -5 + 7\lambda$$

$$y = 2 - 9\lambda$$

$$3 = 2 - 9\lambda$$

$$z = 3 - 9\lambda$$

$$-3 = 3 - 9\lambda$$

A não está em $\overrightarrow{BC}$

$$\textcircled{2} \text{ a) } \vec{AB} = B - A = (-3, 0, 9) - (0, 1, 8) = (-3, -1, 1)$$

$$\left. \begin{array}{l} x = 1+h \\ y = 2+h \\ z = -3h \end{array} \right\} C = (1+h, 2+h, -3h)$$

$$\vec{AC} = C - A = (1+h, 2+h, -3h) - (0, 1, 8) = (1+h, 1+h, -3h-8)$$

$$B(0, 1, 8)$$

$$\vec{AB} \cdot \vec{AC} = (-3, -1, 1) \cdot (1+h, 1+h, -3h-8)$$

$$\vec{AB} \cdot \vec{AC} = (-3-3h, -1-h, -3h-8)$$

$$-7h - 12 = 0$$

$$\begin{array}{l} -7h - 12 \\ h = -\frac{12}{7} \end{array}$$

$$\left(1 + \left(-\frac{12}{7}\right), 2 + \left(-\frac{12}{7}\right), -3 \cdot \left(-\frac{12}{7}\right)\right) \rightarrow \left(\frac{-5}{7}, \frac{2}{7}, \frac{36}{7}\right)$$

$$\vec{BC} = C - B = (1+h, 2+h, -3h) + (-3, 0, 9) = (-2+h, 2+h, -3h+9)$$

$$\vec{AB} \cdot \vec{BC} = (-3, -1, 1) \cdot (-2+h, 2+h, -3h+9)$$

$$-12 - 3h - 2 - h - 9 - 3h = 0$$

$$-23 - 7h = 0$$

$$-7h - 23$$

$$h = -\frac{23}{7}$$

$$5) A) X = (1, 2, 0) + \lambda(1, 1, 0) + \beta(2, 3, -1)$$

$$X = 1 + \lambda + 2\beta$$

$$Y = 2 + \lambda + 3\beta$$

$$Z = -\beta$$

B)

$$AB = B - A = (1, -1, -1) - (1, 1, 0) \rightarrow (0, -2, -1)$$

$$BA = A - B = (1, 1, 0) - (1, -1, -1) \rightarrow (0, 2, 1)$$

↑

$$X = A + \beta \cdot A(0, -2, -1) + \beta(2, 1, 0)$$

$$X = (3, 1, -1) = \alpha(0, -2, -1) + \beta(2, 1, 0)$$

$$X = 3 + 2\beta$$

$$Y = 1 - 2\alpha + \beta$$

$$Z = -1 - \alpha$$

$$C) AB = B - A \rightarrow (2, 1, -2) - (1, 0, 1) = (1, 1, -3)$$

$$AC = C - A \rightarrow (1, -1, 0) - (1, 0, 1) = (0, -1, -1)$$

$$X = A + \lambda AC$$

$$X = (1, 0, 1) + \lambda(0, -1, -1)$$

$$X = 1$$

$$Y = -\lambda$$

$$Z = -\lambda$$

$$X = A + \lambda AB$$

$$X = 1 + \lambda$$

$$Y = \lambda$$

$$Z = -3\lambda$$

$$X = (1, 0, 1) + \lambda(1, 1, -3) \rightarrow$$

$$6) a) A = (9, 1, 0) \rightarrow X - 9, Y + 1, Z$$

$$\vec{U} = (0, 1, 0) \rightarrow$$

$$\vec{V} = (1, 1, 1)$$

$$\text{Det} \begin{pmatrix} X-9 & Y+1 & Z \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} = 0$$

$$(X-9, 1) + (Y+1, 0) + Z, -1 = 0$$

$$X-9 + (-Z) = 0$$

$$X - Z - 9 = 0$$



$$b) \vec{AX} = X - A = X - (1, 0, 1) = (X-1, Y, Z-1)$$

$$\vec{AB} = \vec{B} - \vec{A} = (-1, 0, 1) - (1, 0, 1) = (-2, 0, 0) \quad AX \cdot (AB \times AC) = 0$$

$$\vec{AC} = \vec{C} - \vec{A} = (2, 1, 2) - (1, 0, 1) = (1, 1, 1)$$

$$\begin{pmatrix} X-1 & Y-2 & Z-3 \\ -2 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix}$$

$$X-1 \cdot 0 + -2(Y-2) \cdot 1 + -2(Z-3) = 0$$

$$0 + (-2Y+4) \cdot 1 + -2Z+6 = 0$$

$$0 + 2Y-4 + (-2Z+6) = 0$$

$$2Y - 2Z + 2 = 0$$

c)

$$\vec{AB} = \vec{B} - \vec{A} = (1, -1, -1) - (1, 1, 0) = (0, -2, -1)$$

$$\vec{AX} = X - A = (X, Y, Z) - (1, 1, 0) = (X-1, Y-1, Z)$$

$$\begin{pmatrix} X-1 & Y-1 & Z \\ 0 & -2 & -1 \\ 2 & 1 & 0 \end{pmatrix} \rightarrow$$

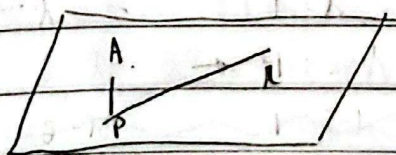
$$(X-1) \cdot 1 + (Y-1) \cdot 1 \cdot 2 + (Z) \cdot 4 = 0$$

$$X-1 + (-2Y+2) + 4Z = 0$$

$$X-1 - 2Y+2 + 4Z = 0$$

$$X - 2Y + 4Z + 1 = 0$$

$$d) X = (0, 2, 2) + \lambda(1, 1, -1)$$



$$\vec{PA} = \vec{A} - \vec{P} = (0, 2, 2) - (1, -1, 1) = (-1, 3, 1)$$

$$X = (1, -1, 1) + \alpha(1, 1, -1) + \beta(1, -3, 1)$$

$$\begin{pmatrix} X-1 & Y+1 & Z-1 \\ 1 & 1 & -1 \\ -1 & 3 & 1 \end{pmatrix}$$

$$(X-1) \cdot 2 + 0 + (Z-1) \cdot 4 = 0$$

$$-2X + 2 + 4Z - 4 = 0$$

$$-2X + 4Z - 2 = 0$$

$$(7) a) 2x + 2y - z + 5 = 0$$

$$A: x=0$$

$$y=0$$

$$z=5$$

$$B: x=0$$

$$y = -2x - 5 \quad y = -5/2$$

$$z=0$$

$$C: x = -7/2, y = -5/2, z = -4/2$$

$$y=0$$

$$z=0$$

$$0$$

$$0$$

$$AB = B - A = (0, -5/2, 0) - (0, 0, 5) = (0, -5/2, -5)$$

$$AC = C - A = (-5/2, 0, 0) - (0, 0, 5) = (-5/2, 0, -5)$$

$$\pi \begin{cases} x = 0 + 0\lambda + (-5/2)\mu \\ y = 0 + (-5/2)\lambda + 0\mu \\ z = 5 + (-5/2)\lambda + (-5)\mu \end{cases}$$

b)

$$5x - y - 1 = 0$$

$$-y = -5x + 1$$

$$+y = 5x - 1$$

$$\pi \begin{cases} x = \alpha \\ y = 5\alpha - 1 \\ z = \beta \end{cases}$$

$$c) z - 3 = 0$$

$$x = \alpha$$

$$y = \beta$$

$$z = 3$$

$$\begin{cases} x = \alpha \\ y = \beta \\ z = 3 \end{cases}$$

$$d) y - z - 2 = 0$$

$$\rightarrow -z = 2 - y \quad z = -2 + y$$

$$x = \alpha$$

$$y = \beta$$

$$z = -2 + y$$

$$\begin{cases} x = \alpha \\ y = \beta \\ z = -2 + \beta \end{cases}$$

$$\textcircled{8} \text{ a) } \begin{cases} x = 1 + \lambda - v \\ y = 2\lambda + u \\ z = 3 - v \end{cases} \quad \begin{aligned} V_1 &= (1, 2, 0) \\ V_2 &= (-1, 1, -1) \\ A &= (1, 0, 3) \\ P &= (x, y, z) \end{aligned}$$

$$AP = P - A = (x, y, z) - (1, 0, 3) = (x-1, y, z-3)$$

$$\begin{array}{c|ccc} x-1 & y & z-3 & \\ \hline 1 & 2 & 0 & (x-1) \cdot (-2) + y \cdot 1 + (z-3) \cdot 3 \\ -1 & 1 & -1 & -2x+2+y+3z-9=0 \\ & & & -2x+y+3z-7=0 \end{array}$$

b)

$$\begin{cases} x = 1 + \lambda \\ y = 2 \\ z = 3 - \lambda + v \end{cases} \quad \begin{aligned} V_1 &= (1, 0, -1) \\ V_2 &= (0, 0, 1) \\ A &= (1, 2, 3) \\ P &= (x, y, z) \end{aligned}$$

$$AP = P - A = (x, y, z) - (1, 2, 3) = (x-1, y-2, z-3)$$

$$\begin{array}{c|ccc} x-1 & y-2 & z-3 & \\ \hline 1 & 0 & -1 & (x-1) \cdot 0 + (y-2) \cdot 1 + (z-3) \cdot 0 \\ 0 & 0 & 1 & -y+2=0 \end{array}$$

c)

$$\begin{cases} x = -2 + \lambda - u \\ y = 2\lambda + 2u \\ z = \lambda + u \end{cases} \quad \begin{aligned} V_1 &= (1, 2, 1) \\ V_2 &= (-1, 2, 1) \\ A &= (-2, 0, 0) \\ P &= (x, y, z) \end{aligned} \quad \begin{cases} AP = P - A \\ \hookrightarrow (x, y, z) - (-2, 0, 0) \\ x+2, y, z \end{cases}$$

$$\begin{array}{c|ccc} x+2 & y & z & \\ \hline 1 & 2 & 1 & (x+2) \cdot 0 + y \cdot (-1) \cdot 2 + z \cdot 2 \\ -1 & 2 & 1 & -2y+2z=0 \end{array}$$

9) a) $\pi: \begin{cases} x = 1 + 2\lambda \\ y = \lambda \\ z = 1 + 3\lambda \end{cases}, \quad S: \begin{cases} x = -1 + \mu \\ y = -1 + 2\mu \\ z = -2 + 6\mu \end{cases}$

$\pi: (1, 0, 1) + \lambda(2, 1, 3)$ $S: (-1, -1, -2) + \mu(1, 2, 6)$

Os vetores diretores são L.D pois $2 \cdot (2, 1, 3) = (4, 2, 6)$.

$-1 = 1 + 2\lambda$

$-1 = \lambda$

$-2 = 1 + 3\lambda$

$\lambda = -1$ Logo são paralelas concidentes.

b) $\pi: X = \overset{A}{(1, 0)} + h(1, 2, 3)$ $S: X = \overset{B}{(2, 3, 3)} + \mu(3, 2, 1)$

Verificar se são paralelos $1, 2, 3 = B(3, 2, 1) = L.I$

Não são paralelos.

Verificar se são concorrentes ou reversas

$AB = B - A \rightarrow (2, 3, 3) - (1, 0) = (1, 2, 3)$

1	2	3	1	2	$6+2+18 = (6+2+18)$
3	2	1	3	2	Det = 0
1	2	3	1	2	

Logo são não paralelos, e são concorrentes

$3, 2, 1 \cdot 1, 2, 3 = 3 \cdot 2 + 3 = 10$

$1 + h = 2 + 3\mu \quad | \quad h = 1$

$1 + 2h = 3 + 2\mu \quad | \quad \mu = 0$

$3h = 3 + \mu$

Ponto Interseção

$(2, 3, 3)$

c) $\begin{cases} x = 2 - 2\lambda \\ y = 2 + 5\lambda \\ z = 11 \end{cases} \quad S: \begin{cases} \frac{x}{2} = \frac{y-1}{-2} = z \\ S: X = (0, 1, 0) + \lambda(2, -2, 0) \end{cases}$

$\pi: X = (2, 2, 11) + \lambda(-2, 5, 0)$

I. Verificar se os vetores diretores são paralelos:

$-2, 5, 0 = B(2, -2, 1) \rightarrow$ são L.I

II. Verificar se são concorrentes ou reversas

$AB = B - A \rightarrow (0, 1, 0) - (2, 2, 11) = (-2, -3, -11)$

-2	-3	-11	-2	-3
-2	5	0	-2	5
2	-2	1	2	-2

$-10 + 0 - 88 = -110 + 0 + 10$
 $-98 - 100 = -200$
 $-200 \neq 0$

Logo são paralelos não reversas

$$d) \pi: \frac{x-2}{3} = \frac{y+2}{2} = z$$

$$\left\{ \begin{array}{l} s: \frac{x}{2} = \frac{y}{2} = \frac{z-3}{2} \\ s: X = (0,0,3) + \lambda(4,2,2) \end{array} \right.$$

$$\pi: X = (2, -2, 0) + \lambda(3, 2, 1)$$

$$s: X = (0, 0, 3) + \lambda(4, 2, 2)$$

1- Verificar se são paralelas: $3, 2, 1 = B(4, 2, 2)$

Não são paralelas, logo é L.I

2- Verificar se são reversas ou concorrentes

$$AB = B - A = (0, 0, 3) - (2, -2, 0) = (-2, 2, 3)$$

(Logo são reversas)

$$\begin{array}{c|c|c} -2 & 2 & 3 \\ \hline 3 & 2 & 1 \\ \hline 2 & 2 & 2 \end{array} \begin{array}{l} -2 \cdot 2 \\ 3 \cdot 2 \\ 2 \cdot 2 \end{array} \begin{array}{l} -16 + 8 + 18 \\ 10 - 56 \\ -46 \neq 0 \end{array}$$

$$(10) a) \pi: \begin{cases} X + 2Y + 3Z - 1 = 0 \\ X - Y + 2Z = 0 \end{cases}$$

1- resolver sistema de equações

$$-X + 2Y - 3Z + 1 = 0$$

$$-3Y - Z + 1 = 0$$

$$\rightarrow Z = 1 - 3Y$$

$$X - Y + 2Z = 0$$

$$-Z = +3Y - 1$$

$$\rightarrow X - Y + (2 \cdot (1 - 3Y)) = 0 \rightarrow X - Y + 2 - 6Y = 0 \rightarrow X = 7Y - 2 \quad Y = T$$

2- Parametriza

$$Z = 1$$

$$X = 7t - 2$$

$$Y = t$$

$$Z = -3t + 1$$

3- Vetorial

$$t=0$$

$$X = -2$$

$$Y = 0$$

$$Z = 1$$

$$(-2, 0, 1) + \lambda(7, 1, -3)$$

$$b) \begin{cases} x + y + z - 1 = 0 \\ x + y - z = 0 \end{cases} \quad y = t$$

1- Resolver Sistema

$$\begin{cases} x + y + z - 1 = 0 \\ x + y - z = 0 \end{cases} \rightarrow \begin{cases} 2x + 2y - 1 = 0 \\ x + y = \frac{1}{2} \end{cases} \quad \begin{cases} x + y + z - 1 = 0 \\ \frac{1}{2} + z - 1 = 0 \end{cases} \rightarrow \begin{cases} z = 1 - \frac{1}{2} \\ z = \frac{1}{2} \end{cases}$$

$$x = \frac{1}{2} - y \rightarrow x = \frac{1}{2} - t$$

2- Parametrizar

$$\begin{aligned} x &= \frac{1}{2} - y & x &= \frac{1}{2} - t \\ y &= t & y &= t \\ z &= \frac{1}{2} & z &= \frac{1}{2} \end{aligned}$$

3- Vetorial

$$\begin{aligned} \text{Se } t=0 & \quad x = \frac{1}{2} \\ & \quad y = 0 \\ & \quad z = \frac{1}{2} \end{aligned} \quad (1/2, 0, 1/2) + \lambda(-1, 1, 0)$$

$$c) \begin{cases} x=3 \\ 2x - z + 1 = 0 \end{cases} \quad \begin{cases} \text{1- Resolver Sistema} \\ x=3 \end{cases} \quad \begin{aligned} x &= 3 \\ y &= 0 \\ z &= 7 \end{aligned}$$

$$2 \cdot 3 - z + 1 = 0 \rightarrow -z = -7 \quad z = 7$$

2- Parametrizar

$$\begin{aligned} x &= 3 \\ y &= t \\ z &= 7 \end{aligned}$$

3- Vetorial

$$\begin{aligned} \text{Se } t=0 & \quad x=3 \\ & \quad y=0 \\ & \quad z=7 \end{aligned} \quad (3, 0, 7) + \lambda(0, 1, 0)$$

$$d) \begin{cases} y=2 \\ z=0 \end{cases} \quad \begin{cases} \text{1- Resolver Sistema} \\ x=t \end{cases} \quad \begin{aligned} x &= t \\ y &= 2 \\ z &= 0 \end{aligned}$$

2- Parametrizar

$$\begin{aligned} x &= t \\ y &= 2 \\ z &= 0 \end{aligned}$$

3- Vetorial

$$\begin{aligned} \text{Se } t=0 & \quad x=0 \\ & \quad y=2 \\ & \quad z=0 \end{aligned} \quad (0, 2, 0) + \lambda(1, 0, 0)$$

① a) $\pi: X = (1, -1, 1) + \lambda(-2, 1, -1)$

$$S: \begin{cases} y + z = 3 \\ x + y - z = 6 \end{cases}$$

1 - Resolver Sistema de S

$$z = 3 - y \quad \rightarrow \quad x + y - 3 + y = 6 \quad y = t$$

$$x + y - (3 - y) = 6 \quad x = 9 - 2y$$

2 - Paramétrica 3 - Vetorial

$$x = 9 - 2t \quad \rightarrow \quad \text{Se } t = 0 \quad x = 9$$

$$y = t \quad y = 0 \quad (9, 0, 3) + \lambda(-2, 1, -1)$$

$$z = 3 - t \quad z = 3$$

Agora podemos estudar as retas

1 - Verificar se elas são paralelas

$$(-2, 1, -1) = \lambda(-2, 1, -1)$$

São Paralelas para $\lambda = 1$

II - Verificar se são distintas ou coincidentes

$$t = 9 - 2t$$

$$-t = t \quad \text{Elas distintas}$$

$$t = 3 - t$$

b) $\pi: \frac{x+1}{2} = \frac{y}{3} = \frac{z+1}{2} \quad \rightarrow \quad X = (-1, 0, 1) + \lambda(2, 3, 2)$

$$S: X = (0, 0, 0) + \lambda(1, 2, 0)$$

1 - Verificar se são paralelas

$$\lambda(1, 2, 0) = (2, 3, 2) \quad \text{são L.I., logo não é paralelo}$$

2 - Verificar se é coincidente ou reversa

$$AB = B - A = (0, 0, 0) - (-1, 0, 1) = (1, 0, 1)$$

$$\begin{array}{c|c} 1 & 0 & 1 & 1 & 0 \\ 2 & 3 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \end{array}$$

$$0 \cdot 10 + 2 \cdot 1 = 13 + 2 + 0$$

$$2 = 15$$

$$3 \neq 0$$

Logo é:
não paralelo
reversa



$$c) X = (8, 1, 9) \wedge \lambda(2, -1, 3) \quad S: X = (3, -2, 4) + \mu(1, -2, 2)$$

1 - Verificar se são paralelas

$$(2, -1, 3) = \lambda(1, -2, 2)$$

Não são paralelas, logo são L.I.

2 - Verificar se é concorrente ou reversa

$$AB: B - A = (3, -2, 4) - (8, 1, 9) = (-5, -5, -5)$$

$$\begin{array}{ccc|cc} -5 & -5 & -5 & -5 & -5 \\ 2 & -1 & 3 & 2 & -1 \\ 1 & -2 & 2 & 1 & -2 \end{array} \quad \begin{array}{l} 10 - 15 + 10 = (-5 + 30 - 20) \\ 5 - 15 \\ -10 \neq 0 \end{array}$$

Logo são reversas

d)

$$\pi: \frac{X+1}{2} = \frac{Y}{1} = \frac{Z}{1} \rightarrow (-1, 0, 0) + \lambda(2, 1, 1)$$

$$S: \begin{cases} X + Y - 3Z = 1 \\ 2X - Y - 2Z = 0 \end{cases}$$

1 - Resolver Sistema

$$3X - 5Z = 1$$

$$X = \frac{1}{3} + \frac{5Z}{3}$$

$$3X = 1 + 5Z$$

$$\frac{1}{3} + \frac{5Z}{3} + Y - 3Z = 1 \rightarrow \frac{5Z}{3} - 3Z + Y = 1 - \frac{1}{3} \rightarrow \frac{5Z - 9Z}{3} + Y = \frac{3 - 1}{3}$$

$$-\frac{4Z}{3} + Y = \frac{2}{3} \quad Y = \frac{2}{3} + \frac{4Z}{3} \quad Z = t$$

2 - Paramétrica

$$X = \frac{1}{3} + \frac{5Z}{3}$$

$$X = \frac{1}{3} + \frac{5t}{3}$$

$$Y = \frac{2}{3} + \frac{4Z}{3}$$

$$\rightarrow Y = \frac{2}{3} + \frac{4t}{3}$$

$$Z = t$$

$$Z = t$$

3 - Vetorial

$$\text{Se } t = 0 \quad X = \frac{1}{3}$$

$$Y = \frac{2}{3}$$

$$Z = 0$$

$$\rightarrow \left(\frac{1}{3}, \frac{2}{3}, 0\right) + \lambda\left(\frac{5}{3}, \frac{4}{3}, 1\right)$$

1- Verificar se são paralelas

$$2, 1, 1 = t \left(\frac{5}{3}, \frac{2}{3}, 1 \right)$$

São L.I., logo não são paralelas

2- Verificar se são concorrentes ou reversas

$$AB = B - A = \left(\frac{1}{3}, \frac{2}{3}, 0 \right) - (-1, 0, 0) \rightarrow \left(\frac{4}{3}, \frac{2}{3}, 0 \right)$$

$\frac{4}{3}$	$\frac{2}{3}$	0	$\frac{4}{3}$	$\frac{2}{3}$	$\frac{4}{3} \cdot 1 \cdot 0 + 0 - (0 + 1 \cdot 0 + 2 \cdot 1)$
2	1	1	2	1	$1 \cdot 1 - 2 \cdot 0$
$\frac{5}{3}$	$\frac{2}{3}$	1	$\frac{5}{3}$	$\frac{2}{3}$	$-6 \neq 0$

Logo são reversas

(12) a) reta

$$r: X = (1, 1, 0) + \lambda(0, 1, 1) \quad | \quad \pi: x - y - z = 2$$

$$(0, 1, 1) \cdot (1, -1, -1) = 0 - 1 - 1 = -2 \neq 0$$

Logo são transversais

• Ponto Intersecção

$$x = 1$$

$$\pi: x - y - z = 2$$

$$y = 1 + \lambda$$

$$1 - (1 + \lambda) - \lambda = 2$$

$$0 + (1 - 2\lambda) = 2$$

$$z = \lambda$$

$$1 - 1 - \lambda - \lambda = 2$$

$$\lambda = -1$$

$$\rightarrow (-1, 0, -1)$$

b)

$$\pi = \frac{x-1}{2} = y = z \rightarrow X = (1, 0, 0) + 2(2, 1, 1)$$

$$X = (3, 0, 1) + \lambda(1, 0, 1) + \mu(2, 2, 0)$$

$X-3$	Y	$Z-1$	$(X-3) \cdot -2 + (Y) \cdot 1 + (Z-1) \cdot 2$
1	0	1	$-2X + 6 + 2Y + 2Z - 2 = 0$
2	2	0	$-2X + 2Y + 2Z + 4 = 0 \div 2$
			$-X + Y + Z + 2 = 0$

$$\rightarrow (-1, 1, 1)$$

$$X = (1, 0, 0) + \lambda(2, 1, 1) \quad \text{e} \quad -X + Y + Z + 2 = 0$$

$$(2, 1, 1) \cdot (-1, 1, 1) \rightarrow -2 + 1 + 1 = 0$$

São Paralelos.

Ver se está contido ou não.

$$-X + Y + Z + 2 = 0 \rightarrow -1 + 0 + 0 + 2 = 0 \rightarrow 1 \neq 0$$

Logo não está contido.

c)

$$\pi: \begin{cases} x - y = 1 \\ x - 2y = 0 \end{cases} \quad \text{1-Resolva Sistema}$$

$$x = 1 + y$$

$$1 + y - 2y = 0 \rightarrow -y = -1 \rightarrow y = 1$$

Parametriza

$$x = 1 + y \rightarrow x = z$$

$$\text{Se } z = 0 \quad x = z$$

$$y = 1$$

$$y = 1$$

$\rightarrow$

$$y = 1$$

$$(2, 1, 0) + \lambda(0, 0, 1)$$

$$z = t$$

$$z = t$$

$$z = 0$$

$$\rightarrow (1, 1, 0)$$

$$(2, 1, 0) + \lambda(0, 0, 1) \quad \text{e} \quad x + y = z$$

$$(0, 0, 1) \cdot (1, 1, 0) = 0, 0, 0$$

12) d) $x - 2y = 3 - 2z$ $x - 3y + 2z = 3$

$$\begin{array}{rcl} x - 2y = 3 - 2z & \rightarrow & -x - 2y + 2z = 0 \\ 3 - 2z + 4 = 2x - 2 & & -2x + 4 - 12 = -3 \end{array}$$

$$\begin{array}{ccc|ccc} 1 & -3 & 2 & 3 & & & \\ & -1 & -2 & 1 & 0 & & \\ & 2 & 1 & -1 & -3 & & \end{array} \begin{array}{l} \\ l_2 \leftarrow l_2 + l_1 \\ l_3 \leftarrow l_3 + 2l_1 \end{array}$$

$$\begin{array}{ccc|ccc} 1 & -3 & 2 & 3 & x5 & 5 & -15 & 10 & 15 & l_1 + l_1 + (-3l_2) & 5 & 0 & 1 & 6 \\ 0 & -5 & 3 & 3 & & 0 & -5 & 3 & 3 & & 0 & -5 & 3 & 3 \\ 0 & -5 & 3 & 3 & l_3 \leftarrow l_3 + l_1 - l_2 & 0 & 0 & 0 & 0 & & 0 & 0 & 0 & 0 \end{array}$$

$$\begin{array}{l} x - 6 = z \\ -5y + 3z = 3 \\ x = 6 - z \\ 5y = 3z - 3 \rightarrow y = \frac{3z - 3}{5} \\ x = 6 - z \\ y = \frac{3z - 3}{5} \\ z = T \end{array}$$

$$P = (1, 2, 0) + \lambda(1, 1, 1) + \mu(2, 1, 0)$$

$$\begin{array}{ccc} x - 1 & y - 2 & z \\ 1 & 1 & 1 \\ 2 & 1 & 0 \end{array} \begin{array}{l} (x-1) - 1 + (y-2) - 2 + z = -1 \\ -x + 1 - 2y + 8 - z = 0 \\ -x - 2y - z + 9 = 0 \end{array} \begin{array}{l} (-1, -2, -1) \end{array}$$

Ver se são transversais

$$(-1, 3/5, 1) \cdot (-1, -2, -1) = 1 - \frac{6}{5} - 1 = 0 \quad \frac{5 - 6 - 5}{5} \neq 0$$

São.

$$\begin{array}{l} -x - 2y - z + 9 = 0 \\ -(6 - T) - 2(3/5T - 3/5) - T + 9 = 0 \\ -6 + T - \frac{6T}{5} + \frac{6}{5} - T + 9 = 0 \\ -\frac{6T}{5} + \frac{31}{5} = 0 \end{array} \begin{array}{l} x = 6 - T \\ y = 3/5T - 3/5 \\ z = T \end{array} \begin{array}{l} x = 6 - \frac{7}{2} \rightarrow 5/2 \\ y = 3/10 \\ z = 7/2 \end{array}$$

$$T = 3/5$$

$$\frac{3}{5} \cdot \frac{7}{2} - \frac{3}{5}$$

$$\frac{9}{10} - \frac{3}{5}$$

$$\frac{9 - 6}{10} = \frac{3}{10}$$

$$\left(\frac{5}{2}, \frac{3}{2}, \frac{7}{2} \right)$$

$$\frac{3}{5} \cdot \frac{7}{2} - \frac{3}{5}$$

$$\frac{21}{10} - \frac{3}{5}$$

$$\frac{21 - 6}{10} = \frac{15}{10} = \frac{3}{2}$$

13) a) $X = (1, 1, 1) + \lambda(2, m, 1)$

$\pi = (0, 0, 0) + \alpha(1, 2, 0) + \beta(1, 0, 1)$

X	Y	Z	$X \cdot 2 + Y \cdot 1 + Z \cdot 2$
1	2	0	$2X - Y - 2Z = 0 \rightarrow (2, -1, -2)$
1	0	1	

$(2, m, 1) \cdot (2, -1, -2) = 4 - m - 2 = 0 \rightarrow 2 = m$

Para que r e π se m tem que ser 2.

b)

$X = (n, 2, 0) + \lambda(2, m, m)$ $\pi: X - 3Y + Z - 1 \rightarrow 1 - 3 + 1$

$(2, m, m) \cdot (1, -3, 1) \rightarrow 2 - 3m + m = 0 \rightarrow 2 = 2m$
 $2 - 2m = 0 \rightarrow 1 = m$

Elas são paralelas com "m" = 1 // Para saber se estão contidas...

$X - 3Y + Z - 1 = 0 \rightarrow (n, 2, 0)$
 $n - 3 \cdot 2 + 0 - 1 = 0$
 $n - 6 + 0 - 1 = 0$
 $n = 7$

$\left. \begin{array}{l} m = 2 \\ n = 7 \end{array} \right\}$

tilibra

$$C) \pi: X = (1, 0, 0) + \lambda(m, 2, m) \quad // \quad x + my + z = 0 \rightarrow (1, m, 1)$$

$$(m, 2, m) \cdot (1, m, 1) = 0 \quad m + 2m + m = 0 \rightarrow m = 0$$

Qualquer valor, m pode assumir. Menos 0

12)

$$a) \pi: X = (2, 2, 1) + \lambda(1, 1, 2) + \mu(3, 3, 1)$$

$X-2$	$Y-2$	$Z-1$	$(X-2) \cdot (-5) + (Y-2) \cdot (-1) + (Z-1) \cdot 0$
1	1	2	$-5X + 20 + 5Y - 10 = 0$
3	3	1	$-5X + 5Y + 10 = 0 \rightarrow (-5, 5, 0)$

$$\pi: X = (3, 0, 0) + \lambda(1, 1, 0) + \mu(0, 1, 2)$$

$X-3$	Y	Z	$(X-3) \cdot 2 + Y \cdot (-1) + Z \cdot 1$
1	1	0	$2X - 12 - Y + Z = 0$
0	1	2	$2X - Y + Z - 12 = 0 \rightarrow (2, -1, 1)$

$(-5, 5, 0) = \lambda(2, -1, 1)$ São L.I. - Transversais e não perpendiculares

Determinar ponto

$$-5X + 5Y = -10 \quad \times 2 \quad -20X + 20Y = -20 \quad \rightarrow \begin{matrix} 5Z & 20 \\ 2 & -1 \end{matrix}$$

$$2X - Y + Z = 12 \quad \times 5 \quad 20X - 20Y + 5Z = 60 \quad \rightarrow \begin{matrix} 5Z & 20 \\ 2 & -1 \end{matrix}$$

$$21X - 21Y + 4Z = 12 \quad 21X - 8Y + 4Z = 4$$

$$X = 2 + 2t \quad X = 2 + 2t$$

$$Y = t \quad Y = t \rightarrow (2, 0, 4) + \lambda(2, 1, 0)$$

$$Z = 4 \quad Z = 4$$

b) $\pi_1: x - y + 2z - 2 = 0 \rightarrow (1, -1, 2)$

$\pi_2: X = (0, 0, 1) + \lambda(1, 0, 3) + \mu(-1, 1, 1)$

$$\begin{vmatrix} x & y & z-1 \\ 1 & 0 & 3 \\ -1 & 1 & 1 \end{vmatrix} \rightarrow \begin{aligned} &x \cdot 3 + y \cdot (-4) + (z-1) \cdot 1 \\ &-3x + (-4y) + (z-1) = 0 \\ &-3x - 4y + z - 1 = 0 \rightarrow (-3, -4, 1) \end{aligned}$$

$(1, -1, 2), (-3, -4, 1)$ são .L.I. Transversais

$(1, -1, 2) \cdot (-3, -4, 1) = -3 + 4 + 2 \neq 0$. São não perpendiculares

• Determinar "reta de intersecção"

$$\begin{aligned} x - y + 2z - 2 &= 0 & \rightarrow & x - y + 2z = 2 \\ -3x - 4y + z - 1 &= 0 & \rightarrow & -3x - 4y + z = 1 \end{aligned} \quad \begin{aligned} x - y + 2z &= 2 \\ 6x + 8y - 2z &= 2 \\ 7x + 9y &= 0 \end{aligned} \quad \begin{aligned} x &= -y \\ 6(-y) + 8y &= 2 \\ 2y &= 2 \end{aligned}$$

$$\begin{aligned} -y - y + 2z &= 2 & \rightarrow & z = y + 1 \\ 2z &= 2 + 2y & \rightarrow & z = 1 + y \end{aligned}$$

$$\begin{aligned} x &= -y & x &= -t \\ y &= t & y &= t \\ z &= y + 1 & z &= t + 1 \end{aligned} \rightarrow (0, 0, 1) + \lambda(-1, 1, 1)$$

c) $\pi_1: 2x - y + z - 1 = 0$ $\pi_2: 4x - 2y + 2z - 9 = 0$

$(2, -1, 1)$

$(2, -2, 2)$

$(2, -2, 2) = 2(2, -1, 1) \rightarrow$ LD com $2 = 2$

Porém não são coincidentes pois

$-9 \neq 2 \cdot (-1)$

$$d) \pi_1 : A = (0, 1, 6), B = (5, 0, 1), C = (2, 0, 0)$$

$$\vec{AX} = X - A \rightarrow (x, y, z) - (0, 1, 6) \rightarrow (x, y-1, z-6)$$

$$\vec{AB} = B - A \rightarrow (5, 0, 1) - (0, 1, 6) \rightarrow (5, -1, -5)$$

$$\vec{AC} = C - A \rightarrow (2, 0, 0) - (0, 1, 6) \rightarrow (2, -1, -6)$$

$$\vec{AX} \cdot (\vec{AB} \times \vec{AC}) = 0$$

x	$y-1$	$z-6$	$x \cdot 1 + (y-1) \cdot (-1) \cdot (-6) + (z-6) \cdot 1$
5	-1	-5	$x + 10y + 10 + z - 6 = 0$
2	-1	-6	$x - 10y + z + 2 = 0$

$$1, -10, 1$$

π_2

$$2x + 10y - 2z - 16 = 0 \div 2 \rightarrow (x + 10y - z - 8 = 0) \rightarrow (1, 10, -1)$$

$$(1, -10, 1) = \lambda (1, 10, -1) \text{ não é L.D, logo é transversal}$$

$$1 - 100 - 1 \neq 0 \text{ logo não é perpendicular}$$

Ponto de Intersecção

$$\begin{aligned} x - 10y + z + 2 &= 0 \\ x + 10y - z - 2 &= 0 \end{aligned}$$

$\rightarrow$

$$\begin{aligned} 2x &= 0 \\ x &= 0 \end{aligned}$$

$$\begin{aligned} 0 + 10y - z - 2 &= 0 \\ 10y - z &= 2 \end{aligned}$$

$$z = 10y - 2$$

$$x = 0$$

$$x = 0$$

$$y = t$$

$$y = t$$

$$(0, 0, 0) + \lambda (0, 1, 10)$$

$$z = 10y - 2$$

$$z = 10t - 2$$

15) a) $\pi_1 = \begin{cases} x = \lambda_1 + 2\mu_1 \\ y = m\lambda_1 \\ z = \lambda_1 + \mu_1 \end{cases} \rightarrow (0,0,0) + \lambda(-1, m, 1) + \mu(2, 0, 1)$

$\pi_2 = \begin{cases} x = 1 + m\lambda_2 + \mu_2 \\ y = 2 + \lambda_2 \\ z = 3 + m\mu_2 \end{cases} \rightarrow (1, 2, 3) + \lambda_2(m, 1, 0) + \mu_2(1, 0, m)$

x	y	z	$(x, m) + y(-3) + z(-2m)$
-1	m	1	$m\lambda + 3\mu - 2m\lambda = 0 \rightarrow (m, 3, -2m)$
2	0	1	

$x-1$	$y-2$	$z-3$	$(x-1) \cdot m - (y-2) \cdot m^2 + (z-3) \cdot (-1)$
m	1	0	$m\lambda - m - m^2\mu - 2m^2 - 2 + 3 = 0$
1	0	m	$m\lambda - m^2\mu - z + 2m^2 - m + 3 = 0 \rightarrow (m, m^2, -1)$

Para provar que dois planos são transversais, basta ser L.I

$$(m, 3, -2m) = \lambda(m, m^2, -1)$$

$$m = \lambda m \rightarrow \lambda = 1$$

$$3 = -\lambda m^2$$

Logo isso prova

$$-2m = -\lambda \rightarrow m = \frac{\lambda}{2}$$

$$b) \pi_1: X = (1, 1, 0) + \lambda(m, 1, 1) + \mu(1, 1, m)$$

$X-1$	$Y-1$	Z	$(X-1) \cdot (1-m-1) + (Y-1) \cdot (1) \cdot (m^2-1) + Z \cdot (m-1)$
m	1	1	$(mX - X) - m + X(-m^2 + 1) + m^2 - 1 + (mZ - Z) = 0$
1	1	m	$X(m-1) + Y(-m^2+1) + Z(m-1) + m^2 - m = 0$

$$(m-1, -m^2, m-1)$$

$$\pi_2: 2x + 3y + 2z + n = 0 \quad (2, 3, 2)$$

Ver se são paralelos

$$\begin{aligned} 2 &= (m-1)B & \frac{2}{m-1} &= \frac{3}{-m^2+1} & 8m-3 &= -2m^2+21 & -2m^2-3m+5 &= 0 \\ 3 &= (-m^2+1)B \end{aligned}$$

$$\begin{aligned} -2m^2 - 3m + 5 &= 0 & +3 \pm \sqrt{9-4 \cdot (-2) \cdot 5} & & \frac{3 \pm \sqrt{49}}{-2} & \rightarrow & \frac{3+7}{-2} = \frac{3-7}{-2} \\ & & & & -\frac{10}{2} = -\frac{5}{1} & & \end{aligned}$$

$$\textcircled{0} \quad m^2 - m = n \rightarrow 1^2 - 1 = n$$

$$\frac{-5 \pm \sqrt{25 - 4 \cdot (-1) \cdot n}}{2} = \frac{25 - 5}{2} = n \quad \frac{25 - 10}{2} = n \quad \frac{15}{2} = n$$

$$\text{Logo } n \text{ e } m \neq 0 \text{ e } 1 \quad // \quad \frac{15}{2} \text{ e } -\frac{5}{2}$$